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Mechanical Engineering, Aerospace Concentration

2023 - 2027



GW Baja SAE
Drivetrain Lead

Baja SAE is a student competition where teams design, build, and race an off-road car

- *Design:* I designed the drivetrain, the system that gets power from the engine to the wheels
- *Manufacturing:* made real parts, and some broke, which taught me more than a textbook
- *Testing:* after every test drive I take the gearbox apart to see what held up and what didn't, and I'm planning to test parts until they fail so I know their real limits
- Leadership: I went from joining the team to leading a subsystem, and anyone can do that if they show up and take ownership
- Takeaway: in a club you do the whole design, build, test, and fix loop that classes only touch on



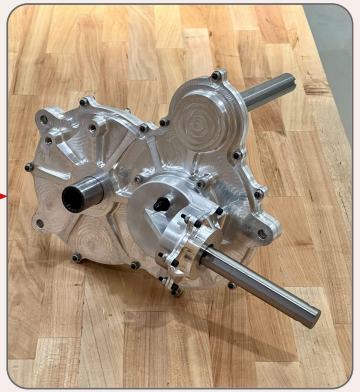




Design



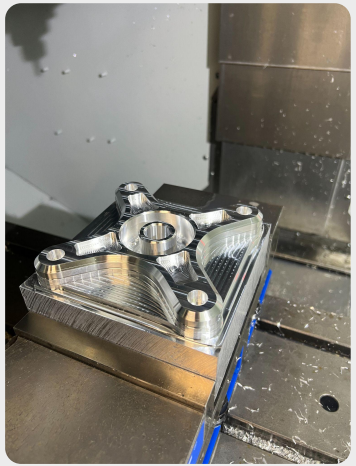
Manufacturing



Testing

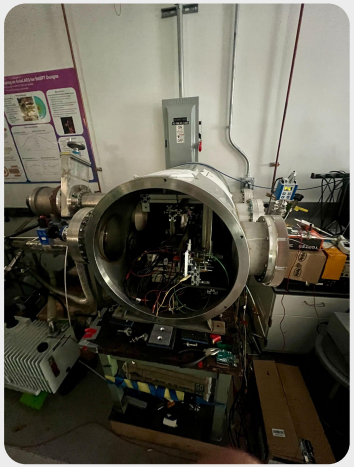
Machine Shop/Research/Capstone

- Building Baja parts led me to the machine shop, where I learned to turn a design into a real part
- Once I could make things myself, I started designing better, because I knew what was actually possible to build
- *Research:* I got into a propulsion lab, building thruster hardware and testing it in a vacuum chamber
- *Capstone:* now I lead the levitation and nozzle systems for our senior project, which uses magnets to hold a small satellite in the air
- Takeaway: spend time in the shop and the lab early, because that's where class material starts to make sense



Machine Shop

Research





Capstone

Boom Supersonic
Build Engineering Intern





Boom is building a supersonic passenger jet, and I helped turn its engine designs into real hardware

- I interned on the build and integration side, helping turn designs into real hardware
- I developed assembly steps for a new engine and wrote code that saved the team hours of manual work
- Seeing how a real aerospace company runs showed me how my classes connect to an actual job
- What got me the internship was what I'd already built and led at Baja, not my grades alone
- Takeaway: interviewers ask what you've owned and built, so start building

